Lithium-Ion Battery Recycling Equipment Guide

Machinery, Brands, and Technical Specifications for Black Mass Production

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1. Introduction

This document provides a comprehensive overview of the equipment required for lithium-ion battery recycling, specifically for black mass production through mechanical recycling processes. The information presented herein is intended to assist facility planners, engineering teams, and procurement specialists in selecting appropriate machinery and understanding the technical rationale behind equipment choices.

The mechanical recycling approach described in this document represents the most economical, fastest, and most proven technology pathway for black mass production. Unlike pyrometallurgical methods that require high-temperature furnaces exceeding 1,400 degrees Celsius, mechanical recycling operates at ambient temperatures using standard industrial equipment. This approach has been successfully deployed at commercial scale by major recyclers worldwide, including Li-Cycle, Redwood Materials, Umicore, and Fortum.

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2. Process Equipment Overview

The following sections detail the recommended equipment for each stage of the lithium-ion battery recycling process, including manufacturer information, technical specifications, and the rationale for each recommendation.

2.1 Primary Shredder

The primary shredding stage represents the first aggressive mechanical processing step in battery recycling, where battery cells are mechanically reduced to fragments suitable for material separation.

Recommended Equipment: Rotary Shear (VR Type) - Inertizable

Specification	Details
Equipment Type	Rotary Shear (VR Type) with Inertizable Design
Key Suppliers	BHS-Sonthofen, MTB, Andritz
Function	Breaks open battery cell casings and releases internal electrode materials under controlled conditions

Technical Description

The primary shredder is a critical piece of equipment in the battery recycling process. It operates under an inert nitrogen atmosphere to prevent fires and thermal events during the shredding operation. The equipment reduces battery materials to pieces of approximately 50-100 millimeters in size. Some facilities employ two-stage shredding, with a coarse primary shred followed by a finer secondary shred.

Why Recommended

- **Gas-tight design**: Prevents oxygen exposure to highly reactive electrode materials
- **Nitrogen-purged operation**: Critical for safety, as freshly exposed electrode materials can ignite if exposed to oxygen
- **Proven at commercial scale**: Successfully deployed at BASF and Li-Cycle facilities
- **Technology redundancy**: Multiple global suppliers ensure supply chain reliability

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2.2 Secondary Crusher

The secondary crushing stage further reduces particle size to liberate valuable materials from current collectors and separators, maximizing black mass yield and purity.

Recommended Equipment: Universal Shredder (NGU Type) or Hammer Mill

Specification	Details
Equipment Type	Universal Shredder (NGU Type) or Hammer Mill
Key Suppliers	BHS-Sonthofen, Stokkermill, Metso
Function	Fine liberation of electrode materials from current collectors and separators with adjustable particle size

Technical Description

Primary shredded material is fed through secondary crushers and mills to achieve finer particle sizes, typically less than 20 millimeters. Hammer mills, impact crushers, or ball mills may be used depending on throughput requirements and desired particle size distribution. The milling process liberates the active electrode materials (cathode and anode powders) from the copper and aluminum current collector foils. Multiple milling stages may be employed with intermediate screening to achieve optimal liberation and particle size distribution.

Why Recommended

- **Fine liberation**: Effectively separates electrode materials from metallic current collectors
- **Adjustable particle size**: Allows customization based on downstream processing requirements
- **Multiple supplier options**: Reduces dependency on single-source equipment providers
- **Proven reliability**: Well-established technology in industrial applications

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2.3 Screening Equipment

Screening equipment is essential for classifying materials by size and preparing them for separation processes.

Recommended Equipment: Vibrating Screen + Rotary Drum

Specification	Details
Equipment Type	Vibrating Screen + Rotary Drum
Key Suppliers	Major Industrial Suppliers
Function	Classifies crushed materials by size, separating fine black mass powder from larger fragments

Technical Description

The crushed material passes through a series of screening operations where vibrating screens classify materials by size. Fine powder (black mass) passes through the screen mesh while larger fragments are recirculated for further milling. Rotary drums provide additional screening capacity and help achieve more uniform particle size distribution.

Why Recommended

- **Standard technology**: Well-established and understood in industrial applications
- **Multiple suppliers**: Reduces procurement risk and ensures competitive pricing
- **Reliable sizing**: Provides consistent separation performance
- **Low maintenance**: Simple operation with well-known maintenance requirements

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2.4 Material Separation Systems

The separation stage is where black mass is isolated from other battery components through multiple physical separation methods.

Recommended Equipment: Magnetic + Eddy Current + Air Classification

Specification	Details
Equipment Type	Magnetic Separators, Eddy Current Separators, Air Classifiers
Key Suppliers	Eriez, Steinert, Metso
Function	Separates different material fractions based on size, density, and magnetic properties

Technical Description

The separation process employs multiple technologies in sequence:

- **Magnetic Separators**: Extract ferrous materials including steel casings and some current collectors
- **Eddy Current Separators**: Recover aluminum and copper components from non-metallic materials
- **Air Classifiers**: Use density differences to further refine material fractions, separating black mass from lighter plastic and separator materials

The black mass, being a fine powder with intermediate density, is collected separately from heavier metal fractions and lighter plastic materials.

Why Recommended

- **Proven technology**: Long track record in recycling and mineral processing industries
- **High recovery rates**: Effectively recovers valuable metals from battery waste
- **Multiple supplier options**: Eriez, Steinert, and Metso are industry leaders
- **Modular design**: Can be configured to meet specific facility requirements

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2.5 Safety Systems

Safety systems are critical for protecting workers, equipment, and the environment during battery recycling operations.

Recommended Equipment: Nitrogen Inertization + Fire Suppression

Specification	Details
Equipment Type	Nitrogen Inertization Systems, Fire Suppression Systems
Key Suppliers	Generon, Fire System Vendors
Function	Prevents thermal runaway, fires, and toxic gas release during processing

Technical Description

Safety systems encompass multiple components:

- **Nitrogen Generation and Supply Systems**: Maintain inert atmosphere in shredding and crushing enclosures
- **Gas Detection and Scrubbing Systems**: Monitor for hydrogen fluoride (HF) and other toxic gases
- **Fire Suppression Systems**: Detect and extinguish fires before they spread
- **Explosion Venting**: Safe release of pressure in case of thermal events

All shredding equipment must be housed in nitrogen-purged enclosures with explosion venting, fire suppression systems, and continuous gas monitoring.

Why Recommended

- **Critical for safe operation**: Prevents thermal events that could endanger workers and damage equipment
- **Prevents thermal runaway**: Lithium-ion batteries can experience thermal runaway if not properly handled
- **Protects investment**: Reduces risk of facility damage and business interruption
- **Regulatory compliance**: Meets occupational safety and environmental requirements

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3. Equipment Summary Table

Process Stage	Recommended Equipment	Key Suppliers	Primary Function
Primary Shredder	Rotary Shear (VR Type) - Inertizable	BHS-Sonthofen, MTB, Andritz	Break open battery cells under nitrogen atmosphere, reduce to 50-100mm fragments

Process Stage	Recommended Equipment	Key Suppliers	Primary Function
Secondary Crusher	Universal Shredder (NGU) or Hammer Mill	BHS-Sonthofen, Stokkermill, Metso	Fine liberation of electrode materials, reduce to <20mm particles
Screening	Vibrating Screen + Rotary Drum	Major Industrial Suppliers	Classify materials by size, isolate black mass fraction
Separation	Magnetic + Eddy Current + Air Classification	Eriez, Steinert, Metso	Separate metals (Fe, Al, Cu) from black mass and plastics
Safety Systems	Nitrogen Inertization + Fire Suppression	Generon, Fire System Vendors	Prevent fires, thermal events, and toxic gas release

4. Technology Redundancy and Supply Chain Benefits

A key advantage of the mechanical recycling approach is the technology redundancy built into the equipment selection. All required equipment—including shredders, screens, magnetic separators, and eddy current separators—is manufactured by multiple global suppliers. This eliminates single-source dependencies and ensures:

- **Competitive pricing**: Multiple suppliers drive competitive pricing
- **Supply chain reliability**: No single point of failure in the supply chain
- **Serviceability**: Well-established maintenance procedures and trained technicians
- **Scalability**: Equipment can be sized to meet various throughput requirements

5. Conclusion

Selecting the appropriate equipment for lithium-ion battery recycling facilities requires careful consideration of safety, efficiency, reliability, and supply chain factors. The equipment recommendations provided in this document represent the best available technology for mechanical black mass production, with proven track records at commercial-scale facilities worldwide.

The combination of rotary shear primary shredders, hammer mill or universal shredder secondary crushers, vibrating screening equipment, magnetic and eddy current separation systems, and comprehensive safety systems provides a robust and reliable processing train for lithium-ion battery recycling operations.

Document prepared for lithium-ion battery recycling facility planning and equipment procurement purposes.